Recommendation Letter for Mr. Valdemar Tsanov

I am writing in support of Mr. Tsanov's application for support for the Summer School on Structures in Lie Representation Theory. Valdemar Tsanov was enrolled in the Ph. D. program at Queen's in the Fall of 2007. He holds a M. Sc. degree in Mathematics from Sofia University. His background is solid with a large number of courses covering most major areas of Pure Mathematics. His master's thesis problem grew into a research project that he completed before his arrival in Kingston. Vardemar studied geometric properties of certain topological objects, the so called torus knots. He has prepared a manuscript of 48 pages containing original results, which I believe will be published in a respected journal.

Valdemar's current research project deals with a natural problem related to the classical Borel-Weil-Bott theorem. Consider a homogeneous variety G/B, where G is a reductive algebraic group and B is a Borel subgroup of G. The Borel-Weil-Bott theorem claims that the cohomologies of line bundles on G/B are irreducible G-modules and, conversely, every irreducible G-module can be realized in this way in every cohomological degree. An embedding of G/B into another homogeneous variety G`/B` induces a natural G-module map from the cohomology of a line bundle on G`/B` to the cohomology of the corresponding line bundle on G/B. It is natural to study this map. Mike Roth and I recently established necessary and sufficient conditions for the surjectivity of this map in the case when G is embedded diagonally in GxG. Our methods are purely geometric and do not seem to carry over to the general case. Valdemar is currently working on extending these results by first establishing functoriality of the identification of cohomoligies on G/B with Lie algebra cohomologies on the nilpotent radical of B and then studying the analog of the original problem for Lie algebra cohomologies. I am happy to report that he has made a significant progress and I expect that soon he will have a complete answer for an arbitrary embedding of algebraic groups.

Future problems for Valdemar include studying the representations that arise from this construction, extending the results to some infinite-dimensional groups, and, possibly, supergroups. I believe that the Summer School in Bremen will help Valdemar prepare for these new directions of research. Indeed, more than half of mini courses planned are related to different aspects of these problems. I hope that he will be given the opportunity to attend and benefit from the excellent program planned for the Summer School.

Ivan Dimitrov

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